



Dual-phase compression–torsion metamaterial for metro track vibration control

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ABSTRACT

This study introduces a dual-phase compression–torsion metamaterial (DP-CTM) concept that integrates load-bearing capability with vibration attenuation for metro track structures. Unlike conventional vibration control approaches that rely on stiffness reduction or local resonance, the proposed DP-CTM enables functional integration through compression–torsion (CT) coupling enabled by the internal skeleton. Under axial loading, the internal compressive–torsional skeleton converts vertical compression into rotational deformation via constrained topology, while the thermoplastic polyurethane (TPU) layer functions as a constrained viscoelastic damping layer, enabling efficient hysteretic energy dissipation through shear-dominated deformation. To achieve reliable design under limited data, a data-efficient sequential surrogate-based multi-objective optimization framework is developed for high-precision parameter tuning. Numerical simulations and experimental tests are performed to validate load-bearing and vibration attenuation performance. Results reveal symmetry-dependent deformation and vibration mitigation characteristics. A CT ratio is used to quantify deformation conversion efficiency, showing stronger torsional activation in the antisymmetric configuration, whereas the symmetric configuration provides improved load-bearing performance with $\sim 2.7\%$ lower compressive deformation at 20 kN. Frequency-sweep tests further confirm effective vibration attenuation in the 103–166 Hz range, with average vibration level reductions of 3.1 dB and 3.5 dB for the antisymmetric and symmetric structures, respectively. These findings support a metamaterial-driven alternative to conventional rubber-based isolation solutions for metro track vibration control.

1. Introduction

Urban rail transit represents the developmental trajectory of modern megacities. With its distinct advantages of high efficiency and low carbon footprint, it has become an indispensable pillar of urban infrastructure [1,2]. However, while bringing significant convenience to daily life, urban rail transit also introduces a series of challenges, particularly vibration issues caused by moving trains [3–5]. These vibrations primarily manifest as low-to-mid frequencies ranging from 0 to 500 Hz, originating from the dynamic interactions between the train wheels and rails. They propagate through infrastructure such as tunnels and bridges into the surrounding environment [6–8]. With the

increasing number of metro lines in operation, rising train density and the growing public demand for high-quality living environment, addressing the associated environmental vibration issues has become particularly urgent [9–11].

Existing vibration isolation measures, including vibration-damping fastenings [12], steel spring floating slabs [13], and ballast isolation pads [14], mainly target mid-to-high frequency vibration attenuation in urban rail systems [15]. These systems achieve isolation by incorporating damping components into the track support structure; however, their effectiveness remains strongly dependent on the global stiffness distribution of the track–foundation system. For example, Gao et al. [16] developed a dual-stiffness fastening system that enhances vibration

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reduction while addressing material aging issues. Aikawa et al. [17] demonstrated that the natural vibration modes and global stiffness distribution of ballasted tracks fundamentally control their dynamic response. These findings indicate that vibration mitigation strategies are inherently constrained by the modal properties of the track structure, limiting broadband performance [16–20]. To improve low-frequency isolation, reducing equivalent support stiffness is commonly adopted. Although this lowers the isolation frequency, it weakens the overall structural integrity of the track and degrades ride comfort. Alternatively, vibration absorption devices such as rail dynamic absorbers and rail dampers [21,22] provide localized resonance-based suppression. However, their narrow-band effectiveness and additional structural complexity hinder practical implementation [23,24]. Consequently, existing strategies involve inherent trade-offs between vibration mitigation and structural integrity, leaving integrated load-bearing and vibration control within limited installation space an unresolved challenge.

Mechanical metamaterials, a branch of metamaterials, originated in the field of electromagnetic materials. The concept was first proposed by the Soviet physicist Victor Veselago in 1968 [25]. They transcend the physical properties of conventional materials, exhibiting unique mechanical, acoustic or magnetic characteristics [26–28], such as negative mass density [29,30], negative stiffness [31,32], negative Poisson's ratio [33,34] or programmability [35]. They found extensive applications in vibration control devices, energy absorption systems and similar [36–38]. In recent years, metamaterial-based vibration mitigation has advanced significantly through bandgap engineering and flexural wave manipulation. Representative contributions from the research group of Muamer Kadic [39–42] have systematically established the theoretical and structural foundations of chiral mechanical metamaterials, elucidating low-frequency Fano resonances and non-reciprocal wave propagation mechanisms. These studies have substantially deepened the understanding of wave–structure interactions and enabled frequency-selective attenuation in architected elastic systems. In parallel, geometry-driven heterogeneous designs have been explored to enhance mechanical performance through structural hierarchy [43]. However, most existing metamaterial approaches primarily focus on wave propagation control, with comparatively limited attention to integrated load-bearing capability under practical engineering constraints.

Based on these advances, a new class of coupled CT metamaterials has recently attracted growing attention as an alternative and complementary strategy. Notably, the coupled CT structures proposed by Frenzel et al. in 2017 [44] feature compressive–torsional coupling capable of converting axial compression into torsional deformation, thereby enabling intrinsic deformation-induced energy conversion [45–47]. Subsequent studies further demonstrated that CT coupling can effectively tailor stiffness, redistribute strain energy, and activate additional deformation modes conducive to vibration mitigation and energy dissipation [48,49]. To investigate the deformation mechanism of coupled CT structures, numerous researchers have conducted theoretical studies. Qi et al. [50] explored the mechanical behavior of chiral structures under large deformations by establishing a theoretical model of three-dimensional chiral structures and conducting experiments. Lin et al. [51] developed a new constitutive model based on the Cosserat rod theory and studied the buckling behavior of chiral structures. Kadic et al. [52] described the deformations of chiral structures from a continuum mechanics perspective. CT metamaterials derive their distinctive functionality from topology-driven microstructural design, which enables axial loading to induce cooperative torsional deformation at the unit-cell level. Recent studies have demonstrated the effectiveness of such coupling mechanisms in vibration suppression and wave control. For example, chiral metamaterial-based nonlinear energy sinks have been developed to enhance torsional energy dissipation [53], while curved-beam compression–twist architectures have been employed to tailor bandgap characteristics in phononic systems [54]. Nested,

bistable, and pre-torsion configurations further enable customized mechanical responses and multi-functional integration [55–58]. These studies demonstrate that geometric design enables the conversion of macroscopic compression into controllable microstructural torsion, offering new pathways for vibration mitigation and energy management.

Owing to this unique deformation mode, CT coupled structures have demonstrated two notable advantages. Firstly, torsional deformation provides an effective pathway for multidimensional energy dissipation, enabling both vertical and lateral excitations to be guided into torsional modes, thereby supporting broadband and omnidirectional vibration attenuation [59–61]. Xu et al. [62] designed a novel syndiotactic chiral metamaterial absorber based on the CT coupling effect, demonstrating favorable low-frequency and broadband energy dissipation, and explored its potential applications to mechanical vibrations. Liu [63] combined the Kresling origami structures with the press-torsion coupling to construct a novel structure for low-frequency vibration isolation, investigating its longitudinal and transverse bandgap characteristics. Secondly, CT coupled architectures exhibit inherent potential for integrating load-bearing and energy dissipation functions. Hao et al. [64] and Montazeri et al. [65] characterized their quasi-static mechanical behavior, and Ou et al. [66] demonstrated their impact mitigation capability, confirming the role of distributed torsional deformation in balancing stiffness and energy absorption. However, most existing studies primarily emphasize deformation mechanisms, impact resistance, or wave control concepts, and comparatively less attention have been paid to their applicability in load-bearing vibration mitigation scenarios with stringent geometric constraints, such as railway track systems. This gap highlights the need for CT metamaterial designs that can simultaneously deliver controllable vibration attenuation within relevant low-frequency ranges and sufficient structural stiffness under sustained service loads.

Based on the CT deformation mechanism, this study introduces a DP-CTM structure into railway engineering to integrate vibration mitigation with load-bearing performance. Unlike existing CT metamaterials that mainly emphasize deformation behavior or wave-control effects, the proposed DP-CTM is designed for sustained service loading under strict geometric and stiffness constraints typical of urban rail systems. A metallic CT skeleton converts vertical compressive loads into torsion-dominated deformation, which is coupled with a TPU layer acting as a constrained viscoelastic medium to dissipate energy through hysteretic shear. This mechanism enables vibration attenuation without relying on global stiffness reduction or narrow-band resonance, thereby preserving macroscopic stiffness and geometric stability. Furthermore, the inherent anisotropy of the topology allows independent tuning of vertical compliance and lateral stability. The concept is implemented through theoretical modeling, data-efficient multi-objective optimization, and experimental validation to establish a practical framework for load-bearing vibration-mitigating track structures.

The paper is organized as follows. Section 2 presents the parametric modeling framework of the DP-CTM and introduces a sequential surrogate-based optimization strategy for the multi-objective co-optimization of vibration mitigation performance and load-bearing capacity. Section 3 describes the experimental investigations conducted under static and dynamic loading conditions to validate the proposed design and its mechanical response. Section 4 analyzes the mechanical behavior and energy dissipation mechanisms of the DP-CTM, with particular emphasis on CT coupling and vibration transmission control. Section 5 discusses representative engineering applications in metro track systems, including vibration-damping fastener system and impact mitigation devices for turnout zones. Finally, Section 6 summarizes the principal findings and outlines potential directions for practical implementation. Appendix A compares power flow transmission loss with vibration level drop to clarify their theoretical differences and applicability. Appendix B presents the Universal Kriging (UK) formulation and implementation details. Appendix C evaluates convergence criteria for adaptive point addition in surrogate-based optimization, while

Appendix D documents the geometric refinement used to mitigate stress concentration and its influence on the global mechanical response of the DP-CTM structure.

2. DP-CTM design and finite element simulation

This section describes the design concept and numerical implementation of the proposed DP-CTM. The structural configuration is first introduced, followed by the evaluation methodology adopted to quantify load-bearing capacity and vibration mitigation performance. The finite element modeling framework is then detailed, including boundary conditions and solution procedures. Finally, a sequential surrogate-based optimization strategy is presented to determine the optimal structural parameters that balance stiffness and vibration reduction requirements.

2.1. DP-CTM geometric model design

This subsection introduces the geometric configuration and assembly layout of the DP-CTM, detailing the arrangement of the compressive-torsional (CT) unit cells and the key structural parameters defining the metamaterial architecture. The dual-phase compressive-torsional metamaterial (DP-CTM) investigated in this study comprises a 3×3 array of compressive-torsional unit cells, as shown in Fig. 1(a). Each unit cell is composed of an external elastic covering layer and an internal coupled CT skeleton, forming a dual-phase load-bearing and energy-dissipative architecture. The elastic cover structure is made of polyurethane (TPU-95A) [67], leveraging its viscoelastic hysteretic behavior to achieve energy consumption and vibration reduction. The internal skeleton structure is made of heat-treated stainless steel (17-4PH) [68, 69], utilizing its coupled CT performance to fulfill load-bearing functions [70]. In Fig. 1(b), the geometry of the DP-CTM skeleton is inspired by S-type metallic and auxetic metamaterial units, which exhibit geometry-induced CT coupling under axial loading; this principle is extended here into a load-bearing skeleton and combined with a viscoelastic encapsulating layer to enhance deformation conversion and vibration dissipation [71,72].

The overall structure has a width and height of W and a depth of D , corresponding to a single cell width and height of d and a depth of w ($W = nw$ and $D = nd$, where n is the number of cells in the array). The coupled CT skeleton structure combines an inner circle and an outer circle, as shown in Fig. 1(c). The outer circle has a radius of r_1 and a wall thickness of $b = r_1 - r_2$, while the inner circle has a radius of r_3 . The internal inclined structure forms an angle α with the vertical direction and has a thickness of c . The elastic covering structure is formed by

rotating and replicating a single covering unit, with an internal radius of r_1 and an inter-unit spacing of t . To optimize vibration damping performance for the CT structure at finite heights, the design variables for each unit cell include r_1 , r_2 , r_3 , α , c and t . The parameter values are listed in Table 1, with w and d fixed at 1 cm, and r_4 fixed at 1.5 mm. This inclined topology is intentionally designed to induce a strong coupling between axial compression and torsional deformation.

It should be noted that the coupling mechanism is intrinsically compression-triggered and arises from a geometry-induced deformation pathway, rather than from a reciprocal response under pure torsional excitation. Consequently, an independently applied torsional excitation does not necessarily activate the same deformation pathway or yield an equivalent energy dissipation mechanism. As a result, the proposed DP-CTM structure exhibits a functionally directional coupling behavior, which is deliberately tailored to compression-dominated service conditions, such as wheel-rail impact excitation. The DP-CTM skeleton and the TPU covering were fabricated separately using polymer fiber laser sintering technology. The skeleton surface was roughened by sand-blasting to enhance interfacial adhesion, and a thin layer of cyanoacrylate adhesive was applied to assemble the two components, forming a stable dual-phase unit for subsequent mechanical analysis. At this stage, the interface between the two phases is assumed to be perfectly bonded, and the potential influence of interfacial degradation is not considered, in order to focus on the dynamic behavior of the proposed dual-phase metamaterial.

To clarify the distinctive working mechanism of the proposed DP-CTM, a concise comparison with representative vibration mitigation strategies, including buckling-based structures, inertia amplification structures, and quasi-zero stiffness (QZS) structures, as shown in Table 2. Buckling-based designs rely on structural instability to achieve stiffness reduction, while inertia amplification structures exploit effective mass enhancement, and QZS systems operate through stiffness cancellation near an equilibrium point. In contrast, the DP-CTM does not depend on instability or inertial effects. Its vibration attenuation mechanism originates from a topology-induced deformation conversion, whereby axial compression is deterministically transformed into torsional deformation. This coupling activates constrained viscoelastic shear in the elastic covering layer, enabling effective energy dissipation

Table 1
Distribution of DP-CTM structural optimization parameters.

r_1/cm	r_2/cm	r_3/cm	$\alpha/^\circ$	t/cm	c/cm
0.35–0.45	0.25–0.35	0.15–0.25	20–70	0.05–0.15	0.05–0.15

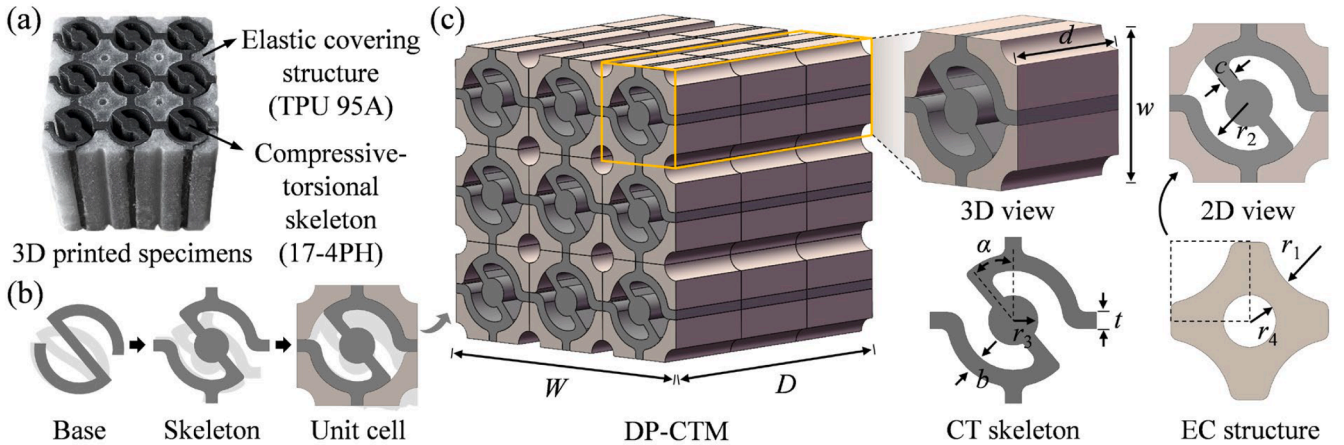


Fig. 1. Structural design of the DP-CTM structure. (a) 3D-printed DP-CTM specimen composed of an elastic covering (TPU 95A) and a CT skeleton (17-4PH); (b) Evolution from the base geometry to the CT skeleton and the corresponding unit cell; (c) 3D and 2D geometric configurations of the DP-CTM structure, including the CT skeleton and elastic covering layer with key geometric parameters.

Table 2

Comparison of vibration mitigation mechanisms relevant to the proposed DP-CTM.

Aspect	Buckling-based structures [51, 64–66]	Inertia amplification structures [49]	Quasi-zero stiffness (QZS) structures [73]	DP-CTM (This work)
Governing mechanism	Structural instability	Effective inertia amplification	Stiffness cancellation	Topology-induced deformation conversion
Dominant deformation mode	Buckling / bending	Translation	Axial deformation	Coupled axial–torsional deformation
Dependence on instability	Required	Not required	Often required	Not required
Primary energy dissipation path	Indirect material damping	Added damping / radiation	Limited material damping	Constrained viscoelastic shear
Load-bearing capability	Degrades after buckling	Moderate	Limited near QZS point	Maintained under compression

while maintaining load-bearing capability under compression.

2.2. Load-bearing and vibration mitigation evaluation method

This subsection establishes an evaluation framework to quantify the load-bearing capacity and vibration mitigation performance of the DP-CTM, considering the installation height constraints associated with railway applications. Structural load-bearing capacity [74] is evaluated by calculating displacement u under 10 kN vertical load during the quasi-static analysis step, while vibration-damping capability is assessed by calculating the power flow drop D_p during the explicit analysis step under 1 kN pulse load.

Regarding vibration reduction system evaluation methods, the force transmission ratio reflects only the dynamic force transfer without accounting for the foundation influence on transmission. Similarly, the vibration level drop fails to fully characterize the system dynamic transmission characteristics because it disregards the impact of effective mass [75]. The power flow [76] method provides an absolute metric for vibration transmission, encompassing both the magnitude of force and velocity as well as their phase relationship, facilitating the understanding of vibration transmission. Appendix A compares the differences between the calculated results of power flow drop and vibration level drop. In a vibration system, the power flow drop is defined as the ratio of power flow above and below the isolation device [77]:

$$D_p = 10 \lg \frac{P_{tr}}{P_{in}}, \quad (1)$$

where P_{tr} represents the power flow transmitted by the isolator, and P_{in} represents the power flow entering the isolator from above. In engineering practice, acceleration measurements are commonly employed, allowing P_{tr} and P_{in} to be expressed as:

$$P_{tr} = -\frac{1}{2\omega} |a_b|^2 \text{Im}\{A_F\}, \quad (2)$$

$$P_{in} = -\frac{1}{2\omega} |a_t|^2 \text{Im}\left\{\frac{A_t^B A_F}{A_t^B + A_F}\right\}, \quad (3)$$

where a_t and a_b represent the acceleration responses above and below the isolator, while A_F and A_t^B denote the apparent masses of the isolator and foundation, respectively.

2.3. Computational model

In this subsection, a unified computational framework is adopted for both the DP-CTM cellular structure and its assembled configuration. Consistent geometric assumptions, material models, and contact formulations are employed, while distinct loading conditions and solution strategies are defined to address different analysis objectives.

For the evaluation of quasi-static mechanical performance, an explicit dynamic approach is adopted to compute the structural response under quasi-static loading, which effectively avoids convergence difficulties associated with severe geometric nonlinearity. For the

assessment of vibration reduction performance, a steady-state dynamic analysis is performed, in which frequency response functions are calculated to characterize the dynamic transmission behavior of the structure. The numerical model consists of three components: an upper rigid platen, a lower rigid platen, and the target structural specimen. Reference points are defined on both platens to facilitate the application of boundary conditions. The upper platen is prescribed with a downward displacement to apply external loading, while the lower platen is fully constrained in all six degrees of freedom. A general contact formulation is adopted, with hard contact governing the normal interaction and a tangential friction coefficient of 0.3 [78]. Binding constraints are applied between different material layers within the specimen to ensure stable interfacial interaction, the details are illustrated in Fig. 2(a).

The upper and lower platens are discretized using rigid shell elements R3D4, which are suitable for representing rigid bodies and improving computational efficiency. The structural specimens are meshed with second-order tetrahedral elements C3D10, which employ modified reduced integration and enhanced shape functions to enhance accuracy in contact-dominated nonlinear analyses. A mesh convergence study is conducted using element sizes of 0.5 mm, 0.6 mm, 0.7 mm, and 0.8 mm. Considering both numerical accuracy and computational efficiency, an element size of 0.6 mm is selected for subsequent analyses.

2.4. Sequential agent model optimization method

This subsection develops a sequential surrogate-based optimization framework to identify the optimal structural parameters of the DP-CTM, enabling coordinated improvement of load-bearing capacity and vibration mitigation performance under prescribed design constraints. Direct uniform sampling would yield a large sample space, but at a substantial computational cost. Additionally, the structure exhibits significant geometric nonlinearity, rendering traditional surrogate modeling methods inaccurate [79]. Recent studies have increasingly explored iterative proxy model updating through dynamic dataset expansion. The variable-fidelity metamodeling method [80] expands low-fidelity datasets by calculating the mean squared error. Kang et al. [81] employed the probability of hypervolume improvement approach to identify new data points for dataset expansion.

The sequential surrogate model optimization is implemented by coupling finite element simulations with a surrogate model. An initial surrogate is constructed using finite element results obtained from a low-fidelity sample set and is iteratively refined by incorporating newly evaluated design points during the optimization process. To ensure numerical reliability, the sampling bounds are strictly controlled to avoid extrapolation, and the optimization is terminated upon convergence [82]. From a methodological perspective, this sequential strategy aims to efficiently exploit limited computational resources by progressively improving surrogate fidelity, rather than relying on a large number of high-fidelity evaluations. By guiding the search toward promising regions of the design space through sequential updates, the optimal solution can be accurately approximated under small-sample conditions.

Fig. 2(b) illustrates the workflow of the sequential surrogate model

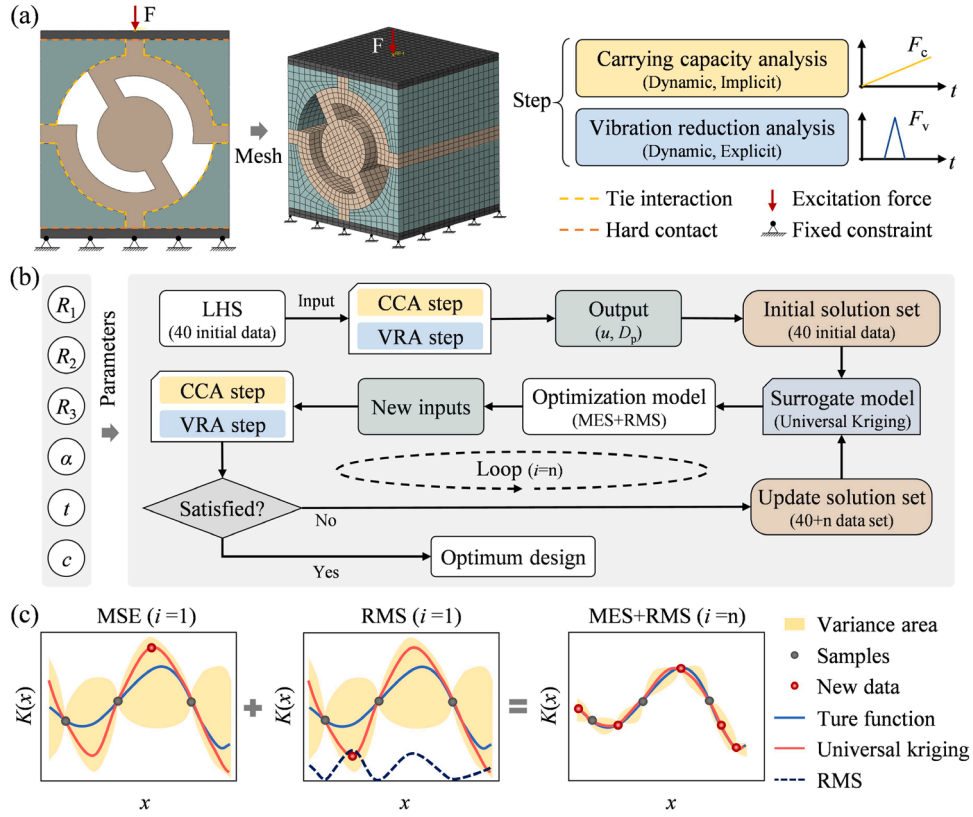


Fig. 2. Finite-element modeling framework and sequential surrogate optimization strategy for the DP-CTM structure. (a) Finite-element model of the DP-CTM unit, including mesh configuration, boundary conditions, and contact definitions, used for carrying capacity and vibration reduction analyses; (b) Sequential surrogate optimization workflow integrating LHS and UK modeling; (c) Adaptive sampling processes under different strategies, including the MSE-based method, the RMS-based method, and MES+RMS combined strategy.

optimization method. A low-fidelity sample set is obtained via Latin hypercube sampling (LHS). Finite element software is then used to find u and D_p , yielding the initial sample solution set. A low-fidelity surrogate model is constructed using the UK. Its estimated values at the sampling points in the design space are expressed as follows:

$$Z^*(\mathbf{X}) = \sum_{i=1}^n w_i(\mathbf{X})Z(\mathbf{X}_i), \quad (4)$$

where n represent the number of sampling points, and $Z(\mathbf{X}_i)$ represents the objective function value at the i th sampling point. The correlation is modeled using a Gaussian kernel function:

$$R(\mathbf{X}_i, \mathbf{X}_j, \boldsymbol{\theta}) = \exp \left[- \sum_{d=1}^D \theta_d (X_{id} - X_{jd})^2 \right], \quad (5)$$

where $\boldsymbol{\theta} = [\theta_1, \theta_2, \dots, \theta_D]^T$ are the correlation coefficients obtained from a maximum likelihood estimation. The detailed UK formulation is provided in [Appendix B](#).

Using the above surrogate model, the MOOOA is employed to find the optimal solutions for displacement u and transmitted power loss D_p [83]. Leveraging the OOA characteristics of fewer parameters and high optimization efficiency [84,85], a Pareto optimal solution archive storage mechanism is introduced to enhance convergence. Solution crowding is calculated to maintain diversity within the Pareto optimal solution set. A hypersphere is defined centered on each trial point. The crowding degree is defined as the number of individuals surrounding the trial point within this hypersphere as follows:

$$\vec{d} = \frac{\vec{f}_{\max} - \vec{f}_{\min}}{n_s}, \quad (6)$$

where $\vec{f}_{\max}$ represents the maximum value vector for each objective function, $\vec{f}_{\min}$ represents the minimum value vector for each objective function, and n_s represents the size of the archive.

The MOOOA framework incorporates a top-level fish-eagle individual selection mechanism for elite trail point identification, together with an adaptive differential t-distribution mutation strategy to strengthen global exploration performance:

$$X_i^{\text{new}} = X_i + (X_{r1} - X_{r2})t_v, \quad (7)$$

where X_i^{new} represents the new trial point generated after the i th fish eagle undergoes mutation, X_{r1} and X_{r2} denote two randomly selected trial points from the population, and t_v is the random variable following a Student's t-distribution with ν degrees of freedom.

The objective function $G(\mathbf{x})$ in the sequential surrogate model is defined as follows:

$$G(\mathbf{x}) = -\beta_1 u(\mathbf{x}) + \beta_2 D_p(\mathbf{x}), \quad (8)$$

where β_1 and β_2 represent the displacement and power transmission loss weighting coefficients, respectively assumed as 0.4 and 0.6. During design optimization, it is desired that $u(\mathbf{x})$ achieves a minimum while $D_p(\mathbf{x})$ reaches a maximum. The objective function is maximized on the surrogate model, subject to the following constraints:

$$\text{s.t.} \begin{cases} \hat{g}_i(\mathbf{x}) \leq 0, & i = 1, 2, \dots, N_c \\ \mathbf{x}_1 \leq \mathbf{x} \leq \mathbf{x}_u \end{cases}, \quad (9)$$

where $\hat{g}_i(\mathbf{x})$ is the surrogate constraint function, N_c is the number of constraints, and $\mathbf{x}_1 \leq \mathbf{x} \leq \mathbf{x}_u$ defines the feasible design domain. Solving the above problem yields new sample points for the low-accuracy

surrogate model (MSP method). The RMSP calculated by the UK model is then subjected to sub-optimization:

$$\begin{cases} \max MSE(G(\mathbf{x})) \\ \mathbf{x}_1 \leq \mathbf{x} \leq \mathbf{x}_u \end{cases}, \quad (10)$$

where $\max MSE(G(\mathbf{x}))$ denotes the surrogate prediction error.

By solving the above problem, the new sample points are obtained. These sample points are then input into the finite element solver. The new sample points are added to the initial solution set, updating the training solution set. By iteratively updating the initial sample point set through this process, a variable-precision surrogate model is constructed. Fig. 2(c) illustrates the methods for acquiring new sample points using MSP and RMS. By combining the point addition techniques, effective fitting between the prediction function and the true function is achieved, yielding a high-precision surrogate model that reflects the load-bearing and vibration-damping performance of the cellular structure. Further details regarding the convergence criteria in the adaptive point-addition strategy are presented in Appendix C.

3. Experimental validation

This section presents the experimental validation of the DP-CTM structure. Uniaxial tensile tests were conducted to determine the constitutive parameters of TPU and stainless steel for calibration of the finite element model. Structural performance was further examined through compression tests, while vibration mitigation capability was evaluated via vibration transmission measurements.

3.1. Material characterization and structural deformation testing

Experimental characterization of the constituent materials and deformation testing of the DP-CTM structure are conducted in this subsection. The obtained material parameters were used to refine the finite element model, while structural compression tests were performed to verify the predicted deformation behavior. To obtain the constitutive parameters of TPU and stainless steel, standard C-shaped dumbbell specimens were fabricated using 3D printing technology in accordance with the ASTM D412 standard [78]. The tensile specimens had a total length of 115 mm, width of 6 mm, thickness of 3 mm and gauge length of 33 mm, as shown in Fig. 3(a-c). Tensile tests were conducted on both the TPU and stainless steel specimens using a TSE-504C universal testing machine. Specimens were clamped and stretched at a rate of 15 mm/min until fracture, after which stress-strain curves were plotted. Engineering stress and strain were converted to true stress and strain, with corrections applied for potential errors caused by the necking effect in the high-strain region.

For the TPU material, a third-order Ogden constitutive model was

employed to capture its nonlinear elastic behavior [86]. The material parameters were identified by fitting the experimental stress-strain curves using the least-squares method and further validated through finite element simulations. For the 17-4PH stainless steel, an elastic-plastic constitutive model was adopted, in which the elastic modulus and Poisson's ratio were defined based on the linear elastic response, while the plastic behavior was described using experimentally measured yield stress-plastic strain data [86]. For clarity and reproducibility, the density, Poisson's ratio, elastic modulus, and constitutive model parameters for both materials adopted in the numerical simulations are explicitly summarized in Table 3. For the DP-CTM structural load bearing performance test, ETM305D microcomputer-controlled electronic universal testing machine was used with a maximum load of 300 kN. Displacement-controlled loading was adopted, with a displacement load range of 0–5 mm, and an initial preload of 0.3 kN was given to obtain an effective displacement-force curve. The loading rate was set to 3 mm/min to simulate quasi-static loading conditions, and the loading results are shown in Fig. 7.

3.2. Vibration reduction performance testing

In this subsection, experimental vibration tests were performed to evaluate the dynamic response and transmission characteristics of the DP-CTM cubic structure. Frequency sweep measurements obtained using the testing system shown in Fig. 4(a) [87] were employed to quantify its vibration reduction capability and to validate the corresponding numerical simulations.

The experimental setup consisted of a signal generator, power amplifier, electrodynamic shaker (Model K2075E, maximum output force 334 N), and a data acquisition (DAQ) system connected to a computer for real-time signal recording and processing. As illustrated in Fig. 4(b), the DP-CTM specimen was rigidly fixed to the vibration table using a steel fixture and high-strength bolts, ensuring a fixed boundary

Table 3

Material parameters of TPU-95A and 17-4PH steel used in the DP-CTM simulations.

Material	Density (kg/m ³)	Poisson's ratio	Elastic Modulus (MPa)	Constitutive model parameters
TPU 95A	1200			Ogden 3N: u/MPa , a (7.627×10^6 , -1.392 ; -25.669×10^6 , 12.499 ; 51.341×10^6 , -24.999)
17-4PH	7800	0.3	19,879.7	Yield stress & Plastic strain (tabulated from experiments)

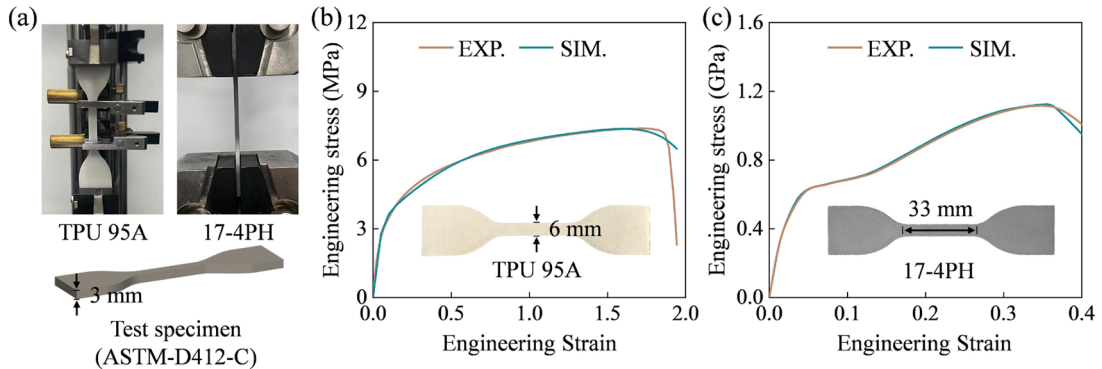


Fig. 3. Experimental setup and constitutive characterization of TPU-95A and 17-4PH materials. (a) Tensile test setups and standard specimens of TPU-95A and 17-4PH; (b) Engineering stress-strain curves of TPU-95A obtained from experiments and simulations; (c) Engineering stress-strain curves of 17-4PH obtained from experiments and simulations.

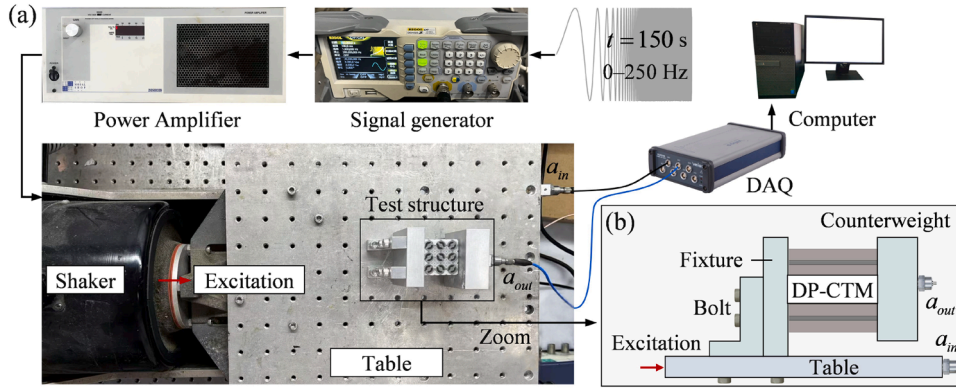


Fig. 4. Experimental setup for vibration reduction testing of the DP-CTM structure. (a) Schematic of the excitation and measurement system, including signal generator, power amplifier, shaker, DAQ, and computer; (b) Detailed test configuration of the DP-CTM specimen, showing the fixture, counterweight, and accelerometer input/output position.

condition during testing. A steel counterweight was mounted on the top surface of the specimen to simulate the attached mass under practical operating conditions. Lateral excitation was applied to the specimen through the shaker–table assembly. Piezoelectric accelerometers (Model INV982X, sensitivity 5 mV/g) were mounted on the vibration table surface and on the top surface of the specimen to measure the input acceleration a_{in} and output acceleration a_{out} , respectively. All acceleration signals were synchronously acquired using a DAQ system (Model INV3062SC) with a sampling frequency of 2048 Hz.

A logarithmic frequency sweep excitation was applied over a frequency range of 1–250 Hz with a constant acceleration amplitude of 1 m/s². The total sweep duration was set to 150 s to ensure sufficient frequency resolution, particularly in the low-frequency range. During the test, the time histories of a_{in} and a_{out} were recorded simultaneously. Fast Fourier Transform (FFT) was then performed to obtain the frequency-domain acceleration responses. The vibration transfer ratio was defined as the magnitude ratio of the output acceleration to the input acceleration in the frequency domain, as expressed in Eq. (11), and was used to quantitatively evaluate the vibration damping performance of the proposed DP-CTM structure:

$$T(f) = \frac{|a_{in}(f)|}{|a_{out}(f)|}, \quad (11)$$

where $a_{in}(f)$ and $a_{out}(f)$ represent the input and output acceleration spectra.

4. Results and discussion

This section presents the analysis of symmetric and antisymmetric configurations constructed from the designed DP-CTM unit according to their symmetry characteristics. The self-contact behavior under quasi-static compression is examined to clarify differences in mechanical responses, with particular emphasis on vertical and lateral stiffness. The vibration performance under harmonic excitation is further compared and interpreted in terms of admittance characteristics.

4.1. Parameter effect analysis and optimization of DP-CTM unit cells

This subsection investigates the parametric influences and coupling mechanisms governing the mechanical and vibration responses of the DP-CTM unit cell, followed by a multi-objective optimization analysis to determine the Pareto-optimal solutions and the most balanced design. For the biphasic pressure-torsion metamaterial unit cell, 40 initial sample points were generated using LHS within the sequential surrogate-based optimization framework described in Section 2.3. Subsequently, 80 iterations were performed, yielding a total of 120

datasets. Parameter analysis was conducted on these datasets, as shown in Fig. 5.

The main effect analysis in Fig. 5(a) indicates that independent variables r_2 and t exert pronounced positive effects on the transmitted power flow difference D_p , suggesting that increasing these parameters enhances the efficiency of CT energy conversion and vibration attenuation. In contrast, the parameters r_1 and r_2 show strong negative effects on the displacement u , reflecting their dominant role in governing the effective stiffness and load-bearing capacity of the unit cell. The main effect plots in Fig. 5(b) and (c) further reveals that both the displacement u and the power flow drop D_p exhibit an inverted U-shaped dependence on the structural angle α . This trend indicates the existence of an optimal angular configuration, at which compression-induced torsional deformation is sufficiently activated to promote energy dissipation without excessively reducing structural stiffness. As the parameter c increases, the displacement u shows a sustained growth trend, implying that larger geometric dimensions facilitate torsional deformation and compliance under compression. The interaction effect analysis in Fig. 5(d) and (e) reveal r_1 increases, displacement u exhibits a sustained growth trend. This trend is more pronounced when t is small. As r_2 increases, the power flow drop D_p continues to grow, and this trend holds when t is large, indicating that favorable geometric combinations can synergistically enhance vibration mitigation performance through CT coupling.

Further multi-objective optimization of the aforementioned dataset was carried out using the MOOQA framework, yielding a set of Pareto-optimal solutions for the biphasic pressure–torsion metamaterial unit cell, as illustrated in Fig. 6(a). The Pareto front explicitly reflects the inherent trade-off between the displacement u under quasi-static compression and the power flow drop D_p under pulse excitation. To facilitate physical interpretation and engineering comparison, four representative solutions (S1–S4) distributed along the Pareto frontier were selected in Fig. 6(b).

Among these candidates, solution S3 corresponds to the optimal compromise determined by the weighting coefficients of displacement and power flow loss defined in Section 2.2. As shown in Fig. 6(b), solution S4 achieves the highest power flow drop D_p , indicating superior vibration attenuation capability. However, this improvement is accompanied by the largest displacement u , implying a pronounced reduction in load-bearing stiffness. In contrast, solutions S1 and S2 exhibit relatively small displacements but provide limited vibration mitigation. Considering the simultaneous requirements of vibration damping performance and structural load-bearing capacity, solution S3 offers the most balanced performance and is therefore identified as the optimal design. The response surface shown in Fig. 6(c) further reveals a highly nonlinear and multimodal relationship among the design parameters c , R_2 , and the resulting power flow drop D_p . Such a complex landscape is characterized by multiple local extrema, which poses

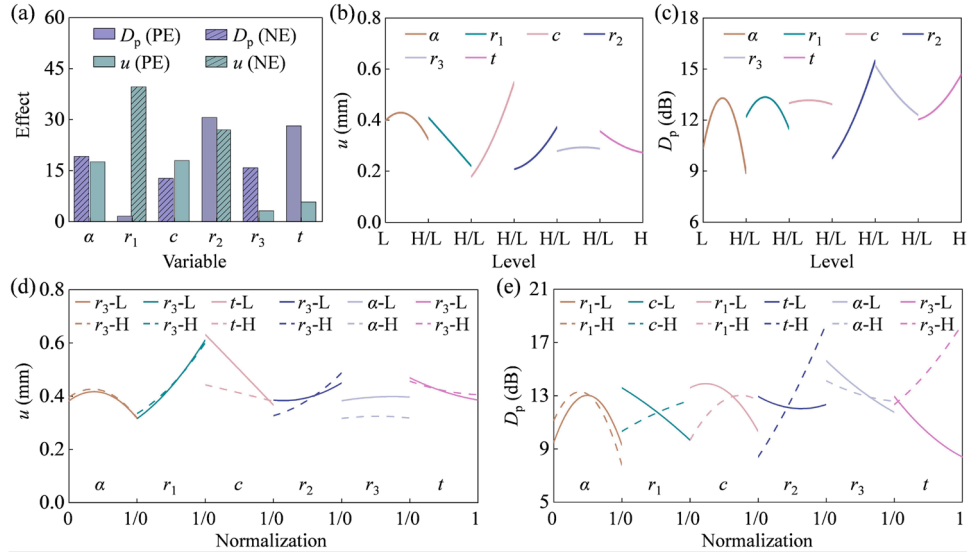


Fig. 5. Parameter effect analysis of the DP-CTM structure. (a) Main effects of design variables on the displacement u and power flow drop D_p ; (b) Influence of individual variables on displacement u across low-high levels; (c) Influence of individual variables on power flow drop D_p across low-high levels; (d) Interaction effects of key variables on displacement u under normalized parameter variations; (e) Interaction effects of key variables on power flow drop D_p under normalized parameter variations. PE and NE denote positive and negative influences, respectively; L and H denote the low and high levels of the variables.

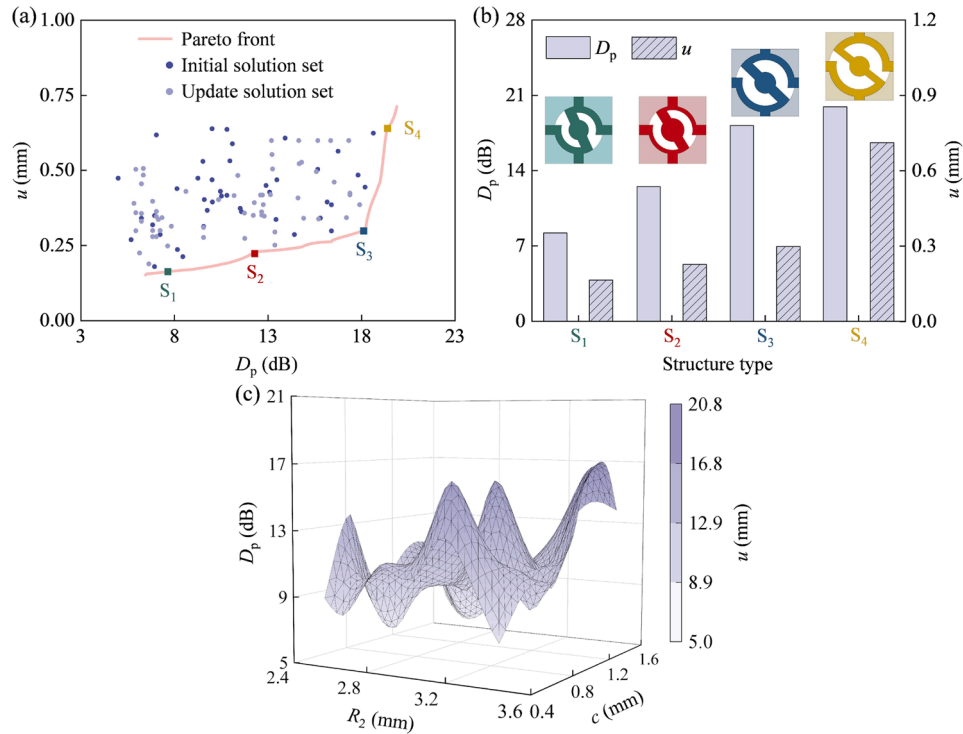


Fig. 6. Multi-objective optimization results and representative DP-CTM designs. (a) Pareto front between displacement u and power flow drop D_p , showing the initial and updated solution sets and selected representative solutions S1-S4; (b) Configurations of the representative DP-CTM structures (S1-S4) and comparison of power flow drop D_p and displacement u ; (c) Response surface of D_p as a function of key design parameters c and R_2 .

significant challenges for conventional gradient-based optimization algorithms [88]. This observation further substantiates the suitability of the adopted MOOAA strategy for exploring the global Pareto-optimal solutions of the DP-CTM unit cell design problem.

4.2. Experimental results and finite element validation

This subsection presents the experimental validation of the DP-CTM

structure. Uniaxial compression tests are conducted to verify its load-bearing capacity and to analyze the deformation characteristics at representative displacement levels. In addition, vibration table tests are performed to evaluate the vibration reduction performance of the structure.

The displacement-force relationship and deformation behaviour of the DP-CTM structure under different displacement loads are verified by comparing the finite element results with the experimental

measurements. Fig. 7 shows the deformation characteristics of the antisymmetric and symmetric structures after loading, respectively, where the load ranges from 0 mm to 5 mm. From the distribution of the overall displacement-force curves, the finite element simulation results are in good agreement with the experimental results, and the difference between the antisymmetric structure simulation and the experimental calculations is 0.1 kN, and 0.5 % under the displacement load of 5 mm. The difference between the symmetric structure simulation and the theoretical calculation results is 0.04 kN, 0.1 %. Comparing the simulation and test results of the antisymmetric structure in Fig. 7(a) and (c), under the condition of 5 mm, the antisymmetric structure has the same deformation trend to the left side, and at the same time, the internal contact order of the structure is the same, which shows the accuracy of the model in simulating the torsional deformation of the compression and torsion structure, and the maximal stress of the Mises stress is 871 MPa. Comparing the simulation and test results of the symmetric structure in Fig. 7(b) and (d), under the same 5 mm displacement loading condition, the deformation characteristics of the internal skeleton have better consistency, in which the contact order of the structure is expanded from diagonal to internal, and the maximum Mises stress is 553 MPa.

Fig. 8(a-b) compares the simulated and experimental vibration reduction results for the aforementioned DP-CTM structure. From the

distribution of peak frequencies, both exhibit good consistency within the primary control frequency bands. Specifically, both the antisymmetric structure simulation and test results exhibit peak frequencies near 113 Hz and 167 Hz, while the symmetric structure simulation and test results show peak frequencies near 82 Hz and 151 Hz. This confirms the validity of the constructed model. In addition, the symmetric structure achieves an average vibration reduction of 3.1 dB, whereas the antisymmetric structure exhibits a higher average reduction of 3.5 dB, demonstrating the superior vibration attenuation performance of the antisymmetric configuration. Some deviation exists in the overall amplitude of the transmission ratio. This primarily stems from the idealized assumptions regarding boundary constraints and load input in the simulation, which differ from the actual table top flexibility, fixture energy dissipation, excitation and constraints adopted in the experiment. The consistent attenuation patterns in key frequency bands between the test and simulation results indicate that the simulation outcomes can serve as a reliable reference for performance evaluation and parameter optimization.

4.3. Load-bearing performance analysis

Building upon the optimized structural design, this subsection systematically investigates the load-bearing behavior of the DP-CTM, with

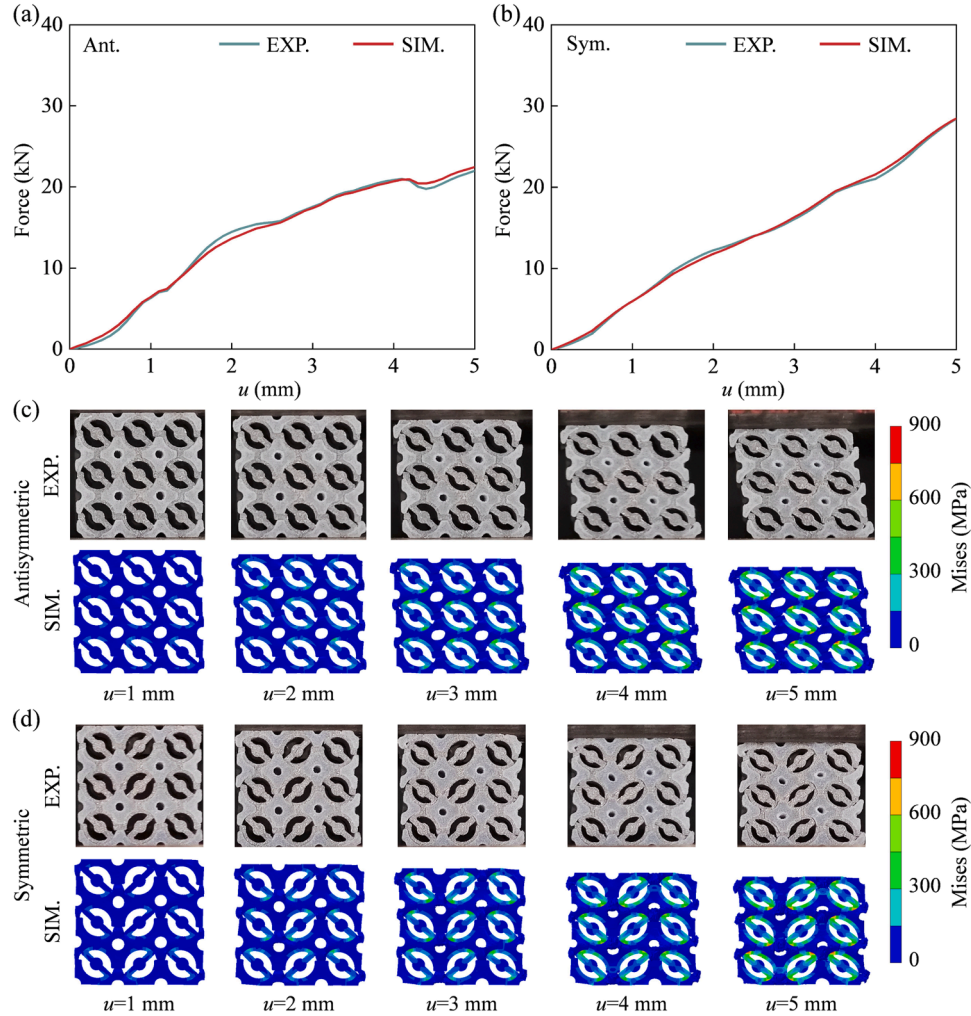


Fig. 7. Experimental validation and deformation evolution of symmetric and antisymmetric DP-CTM structures. (a) Comparison of experimental and simulated force–displacement responses for the antisymmetric DP-CTM structure; (b) Comparison of experimental and simulated force–displacement responses for the symmetric DP-CTM structure; (c) Experimental observations and corresponding simulated von Mises stress distributions of the antisymmetric DP-CTM structure at representative displacement; (d) Experimental observations and corresponding simulated von Mises stress distributions of the symmetric DP-CTM structure at representative displacement.

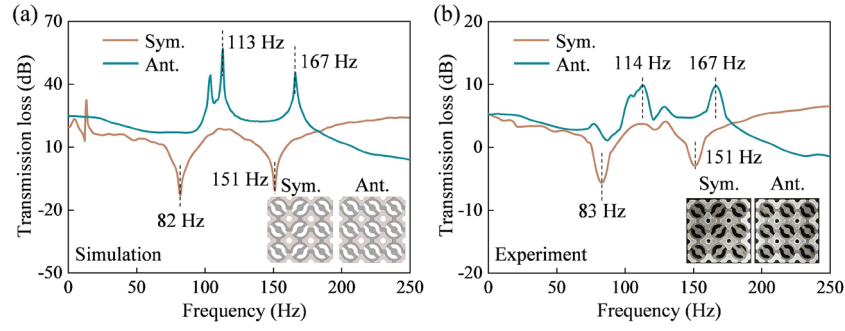


Fig. 8. Comparison of simulated and experimental transmission loss of symmetric and antisymmetric DP-CTM structures. (a) Simulated transmission loss of symmetric and antisymmetric DP-CTM structures; (b) Experimental transmission loss of symmetric and antisymmetric DP-CTM structures.

emphasis on symmetry-dependent CT coupling, stiffness evolution, energy absorption performance, and associated deformation

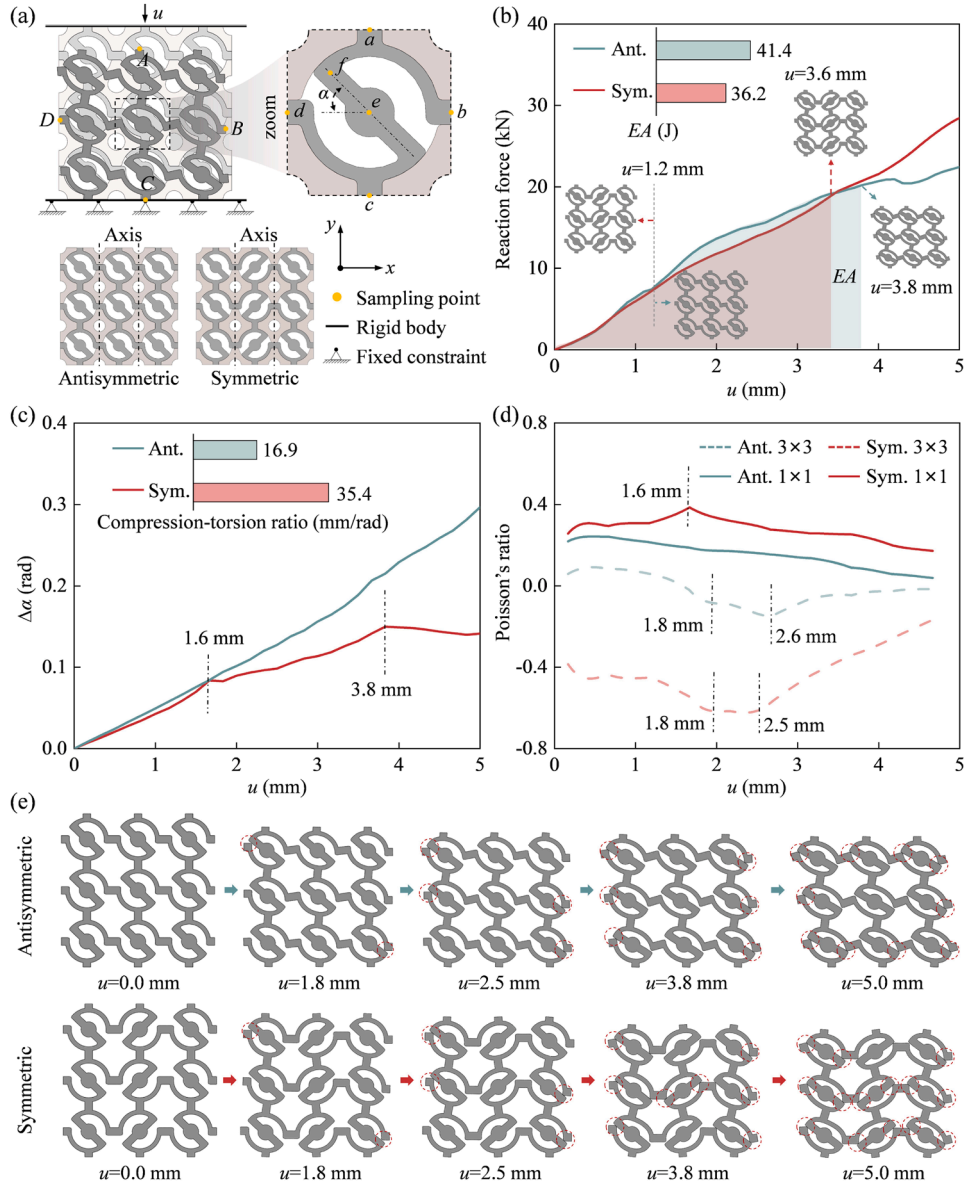


Fig. 9. Mechanical behavior of symmetric and antisymmetric DP-CTM structures. (a) Geometry, loading condition, and sampling points of symmetric and antisymmetric DP-CTM configurations under compression; (b) Reaction force–displacement responses and corresponding deformation energy absorption at different compression levels; (c) Evolution of rotational angle increment $\Delta\alpha$ with displacement u and comparison of CT ratios between symmetric and antisymmetric structures; (d) Poisson's ratio evolution of symmetric and antisymmetric DP-CTM structures under normalized compression; (e) Structure deformation evolution of symmetric and antisymmetric DP-CTM structures at representative compression states.

mechanisms. Based on the S_3 -shaped unit cell structure depicted in Fig. 6, a cubic finite element model with overall dimensions $3\text{ cm} \times 3\text{ cm} \times 3\text{ cm}$ was constructed. Depending on the symmetry, it was categorized into antisymmetric and symmetric structures. To mitigate local stress concentration at sharp corners and improve structural durability, minor geometric refinements including localized chamfering and outer openings were introduced. These modifications primarily redistribute localized stresses without altering the global load–displacement response or stiffness characteristics. Detailed geometric configurations and comparative results are provided in Appendix D.

In the static analysis model, the structure base is fixed, while a 5-mm normal displacement is applied at the top at a loading rate of 1 m/s, as shown in Fig. 9(a). As seen from the displacement-force curve in Fig. 9(b), during the 0–6.9 kN loading phase ($u = 1.2\text{ mm}$), the displacements of the symmetric and antisymmetric structures remain largely consistent, corresponding to the initial contact stage depicted in Fig. 9(e). However, as the load increases (after 3.6 mm), the deformation difference between the two symmetry modes becomes increasingly pronounced. The symmetric structural stiffness is 5.6 kN/mm and the antisymmetric structural stiffness is 5.4 kN/mm in the load range of 0–3.6 mm. Whereas in the range of 3.6–5 mm, the difference between the two stiffnesses is 4.2 kN/mm. This disparity primarily stems from the differing contact sequences of the two structural modes under loading, as illustrated in Fig. 9(e). The energy absorption capacity was quantified by the external work [89,90], defined as:

$$EA = \int_0^u F(x)dx, \quad (12)$$

where $F(x)$ is the reaction force and u is the displacement corresponding to a load of 20 kN.

Under loads ranging from 0 to 20 kN, the antisymmetric structure absorbed 41.4 kJ of energy, representing an increase of 5.2 kJ compared to the symmetric structure. The antisymmetric configuration exhibits a higher propensity for local torsional deformation, resulting in lower stiffness and enhanced energy absorption capacity. Fig. 9(c) illustrates the variation of the DP-CTM structure torsion angle $\Delta\alpha$ with displacement u under loads ranging from 0 to 5 mm. To characterize the CT coupling behavior, the CT ratio is defined as:

$$R_{ct} = \frac{u}{\Delta\alpha}, \quad (13)$$

where the smaller R_{ct} values indicate stronger torsional activation and enhanced energy dissipation capacity.

The CT structure fundamentally absorbs energy by converting compression into local torsional deformation. In the 0–1.6 mm displacement range, both symmetric configurations $\Delta\alpha$ exhibit essentially consistent behavior. However, as the degree of torsional deformation increases, the $\Delta\alpha$ disparity becomes increasingly pronounced. Under a 5 mm displacement load, the antisymmetric Structure $\Delta\alpha$ reaches 0.3 rad while the symmetric Structure $\Delta\alpha$ reaches 0.14 rad, R_{ct} differing by 18.5 mm/rad. Fig. 9(d) illustrates the variation in Poisson's ratio for the internal unit cell structure (1×1 structure) and the composite structure (3×3 structure) with different symmetry configurations. The calculation formulas are as follows:

$$\nu_{1 \times 1} = \frac{\varepsilon_{bd}}{\varepsilon_{ac}}, \quad (14)$$

$$\nu_{3 \times 3} = \frac{\varepsilon_{BD}}{\varepsilon_{AC}}, \quad (15)$$

where ε_{ac} and ε_{AC} denote the applied axial strain of the 1×1 and 3×3 structures, respectively, ε_{bd} and ε_{BD} represent the corresponding transverse strains.

Comparing the deformation of the central cell structure with that of the composite structure reveals that the negative Poisson's ratio is more

pronounced in the 3×3 composite structure. As the displacement load increases, the Poisson's ratios of the positive-symmetric and anti-symmetric structures first decrease and then increase. The minimum Poisson's ratio for the antisymmetric structure is -0.15 , corresponding to a displacement of 2.6 mm, while the minimum for the symmetric structure is -0.63 at a displacement of 2.5 mm. Under a 5 mm displacement load, the symmetric structure Poisson's ratio is 0.15 lower than that of the antisymmetric structure, further demonstrating the antisymmetric structure superior energy absorption capability. Fig. 9(e) analyzes the deformation patterns of the DP-CTM structure under the aforementioned critical displacement loads. The deformation differences between symmetric modes primarily manifest in structural torsional deformation characteristics. Under displacement loading, the whole antisymmetric structure tilts leftward. Self-contact first occurs internally at the upper-left and lower-right corners, then gradually propagates outward. At a 5-mm displacement load, self-contact occurs around the entire structure. Under displacement loading, the symmetric structure exhibits an internal movement where the left side moves downward and the right side upward. At a 5-mm displacement load, self-contact occurs both around the periphery and internally, providing greater stability. This further explains the higher stiffness of this structure.

To investigate the influence of loading rate on the mechanical response of the DP-CTM cube structure, displacement-force curves at 1 m/s, 10 m/s, 20 m/s and 50 m/s were compared, as shown in Fig. 10. As illustrated in Fig. 10(a-b), the overall morphology of the displacement-force curve gradually evolves from quasi-static characteristics to pronounced dynamic features as the loading rate increases. Under loading conditions ranging from 1 to 20 m/s, the curve exhibits a relatively gentle slope. The energy absorption process within the material primarily relies on the CT coupling and self-contacting of structural units.

However, under the high-speed loading with 50 m/s, the peak stress of the curve significantly increases, transitioning from tensile expansion to conventional deformation. As shown in Fig. 10(c-e), under the high strain rate loading, inertial effects suppress the development of compression-induced local torsional deformation, causing the EA at 20 kN to decreased by a factor of approximately 4. From 1 m/s to 50 m/s loading rates, the symmetric structure Poisson's ratio increases from -0.17 to 0.1 , indicating the loss of the designed CT deformation mode. Currently, this structure is primarily applied in metro tracks, where loading rates typically remain below 30 m/s. To mitigate the degradation of CT coupling under higher impact rates, future work will focus on the design of microstructures with reduced rate sensitivity, thereby extending the applicability of the proposed concept to high-speed rail systems.

Further investigation into the coupled vertical and lateral mechanical behavior of the DP-CTM structure was carried out to evaluate its applicability under combined loading scenarios. Simultaneous vertical and lateral displacements up to 5 mm were applied to the DP-CTM structure, and the corresponding force-displacement responses and lateral-to-vertical stiffness ratios for different symmetry configurations are presented in Fig. 11. As shown in Fig. 11(a-b), under initial displacement levels ranging from 0 to 2.2 mm, the lateral reaction force of the symmetric configuration remains relatively low, approximately 1 kN, and then gradually increases to 5.9 kN with further displacement. In contrast, the antisymmetric configuration exhibits a progressive increase in lateral force from 0 to 5.6 kN over the same displacement range, indicating a more effective coupling between vertical compression and lateral deformation.

This difference reflects the distinct deformation pathways induced by the two symmetry modes, with the antisymmetric structure promoting torsion-dominated deformation that more readily transfers vertical loading into lateral response. The resulting lateral-to-vertical stiffness ratios, shown in Fig. 11(c), further highlight this symmetry-dependent anisotropy. Throughout the entire displacement range, the

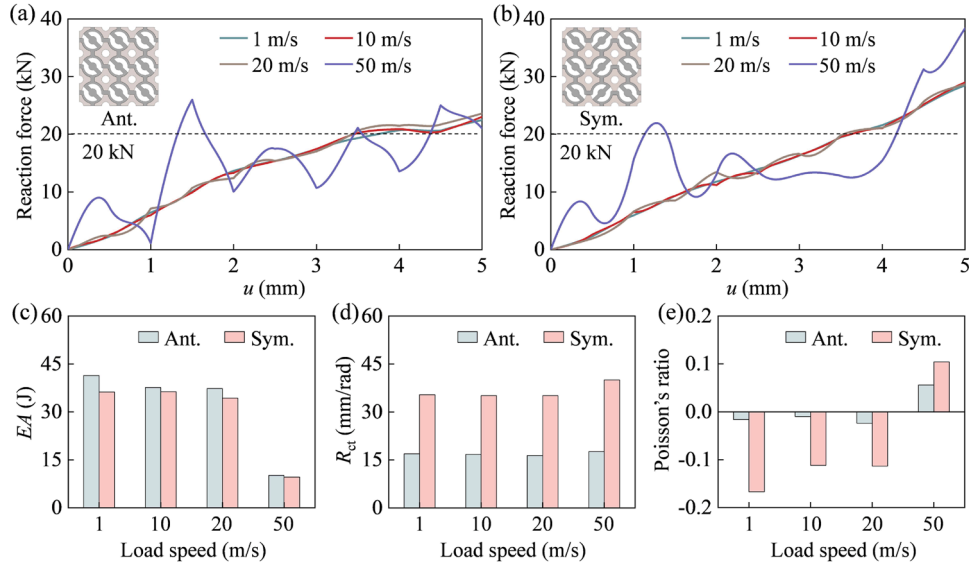


Fig. 10. Effect of loading speed on the mechanical and CT responses of symmetric and antisymmetric DP-CTM structures. (a) Reaction force–displacement responses of the antisymmetric DP-CTM structure under different loading speeds; (b) Reaction force–displacement responses of the symmetric DP-CTM structure under different loading speeds; (c) Comparison of energy absorption under different loading speeds; (d) Comparison of CT ratio R_{ct} under different loading speeds; (e) Comparison of Poisson's ratio under different loading speeds.

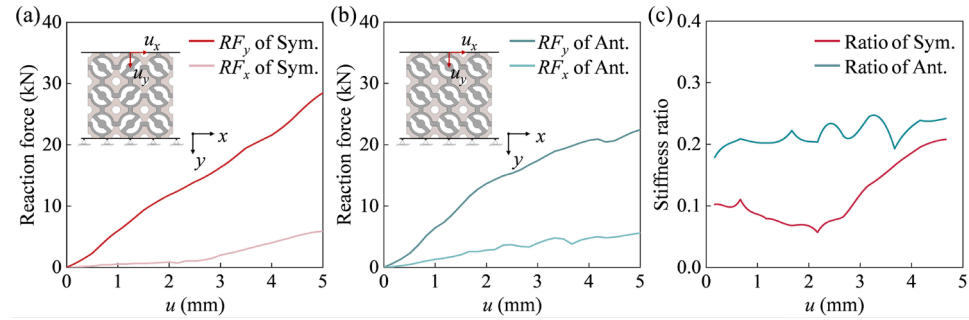


Fig. 11. Evolution of force components and horizontal–vertical stiffness ratio of symmetric and antisymmetric DP-CTM structures. (a) Vertical and horizontal reaction force components of the symmetric DP-CTM structure as functions of displacement; (b) Vertical and horizontal reaction force components of the antisymmetric DP-CTM structure as functions of displacement; (c) Horizontal-to-vertical stiffness ratio for symmetric and antisymmetric DP-CTM structures as functions of displacement.

antisymmetric structure consistently maintains a higher stiffness ratio than the symmetric structure. Notably, the symmetric configuration reaches a minimum lateral-to-vertical stiffness ratio of 0.06 at a displacement of approximately 2.2 mm, demonstrating pronounced directional stiffness decoupling at this stage. Such behavior indicates strong mechanical anisotropy, which is particularly advantageous for engineering applications requiring high vertical load-bearing capacity combined with reduced lateral stiffness.

4.4. Vibration reduction performance analysis

This subsection compares the vibration reduction performance of DP-CTM structures with different symmetry configurations. Harmonic response analysis (HRA) is performed to quantify their vibration mitigation capability, with the underlying differences interpreted through vibration acceleration admittance. In the HRA model, rigid plates are used as constraint layers at both the top and bottom ends of the structure. The load is applied as a 0–250 Hz harmonic excitation force of magnitude 5 kN at the top end, while the bottom constraint layer employs a fixed constraint, as shown in Fig. 12(a). By calculating the input acceleration, output acceleration, and force, the structural power flow transfer difference is determined using Eq. (1). Here, the primary

vibration frequency range of 0–250 Hz in the track is selected as the target frequency band for vibration reduction analysis [91].

Fig. 12(b) compares the power flow loss differences for DP-CTM structures with different symmetry configurations. Overall distribution patterns indicate that the antisymmetric structure exhibits a larger power flow loss difference than the symmetric structure, indirectly demonstrating superior vibration reduction performance for the antisymmetric structure. The symmetric structure shows troughs in its power flow loss difference near 82 Hz and 150 Hz, reflecting structural resonance amplification. The power flow loss of the antisymmetric structure exhibits peaks at 103 Hz, 113 Hz, and 166 Hz, indicating significant vibration attenuation at these frequencies. Admittance characterizes the local excitation–response relationship, whereas transmission loss describes the attenuation of vibration energy during transmission; the two are therefore closely related.

To further elucidate the internal vibration transmission mechanisms, acceleration admittance responses at characteristic points B, C, and D are analyzed, as shown in Fig. 12(c–d). Acceleration admittance characterizes the local excitation–response behavior, while power flow loss reflects the overall wave propagation and energy transmission within the structure; the two are therefore closely related. For the symmetric configuration, within the low-frequency range of 0–50 Hz, the

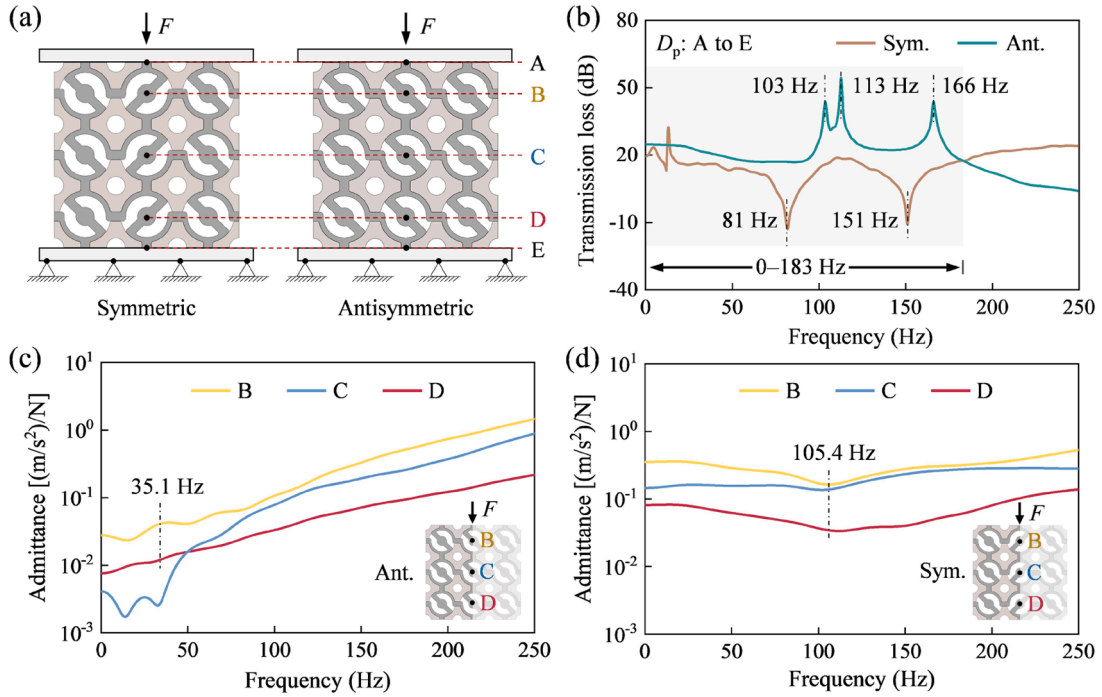


Fig. 12. Vibration transmission characteristics of symmetric and antisymmetric DP-CTM structures. (a) Structural configurations and measurement locations (A–E) of symmetric and antisymmetric DP-CTM specimens under vertical excitation; (b) Power flow loss between locations A and E in the frequency range of 0–250 Hz; (c) Acceleration admittance at locations B–D for the antisymmetric DP-CTM structure; (d) Acceleration admittance at locations B–D for the symmetric DP-CTM structure.

acceleration admittance at point C is consistently lower than that at point D. This behavior arises because the geometric symmetry restricts the effective transmission of excitation energy into the central circular region, rendering it relatively insensitive to low-frequency vibrations. In contrast, the antisymmetric structure exhibits a markedly different admittance distribution. Within the frequency range of approximately 105–200 Hz, the acceleration admittance at point C is significantly higher than that at point D. This pronounced reduction in admittance during wave propagation from C to D indicates rapid internal vibration attenuation, which directly explains the substantial power flow loss observed for the antisymmetric structure in the 100–200 Hz frequency band. These results demonstrate that symmetry-induced differences in internal vibration transmission pathways play a critical role in determining the vibration mitigation performance of the DP-CTM structure.

5. Applications of vibration reduction in track structures

This section evaluates the applicability of the DP-CTM concept in track vibration reduction systems. Through integration into fastening components and frog impact mitigation devices, its CT coupling and stiffness anisotropy are shown to enhance load-bearing stability, vertical-lateral decoupling, and vibration attenuation compared with conventional solutions, thereby demonstrating its engineering potential.

In the field of vibration reduction of metro track structures, existing measures such as vibration-damping fastenings and steel spring floating slabs have demonstrated effectiveness in engineering practice, yet limitations persist in coordinated load-bearing–damping design and vertical–lateral stiffness decoupling. Motivated by these challenges, the DP-CTM structure is introduced as a proof-of-concept mechanical architecture to explore the integration of load-bearing and vibration reduction within a unified metamaterial-inspired framework. The geometry and material selection adopted herein are intended to elucidate the underlying vibration control mechanisms and anisotropic stiffness characteristics rather than to represent a finalized manufacturing solution. On this basis, the application potential of DP-CTM concepts in track

vibration-damping fasteners and frog–wheel impact suppression dampers is discussed, and their advantages for track vibration control are systematically analyzed, while further engineering adaptation and techno-economic assessment are left for future studies.

As a critical component of the track structure, the fastening system not only constrains the rail but also bears and transmits wheel-rail interactions during train operation. The fastening support system exhibits strong nonlinear characteristics [92]. The Vanguard fastener supports the rail at the web, forming a “suspended” structure that significantly enhances the track system isolation of low-to-mid-frequency vibrations. In existing lateral shoulder designs, rubber materials predominate. Under train loads, the shoulder controls vertical loads through vertical shear deformation and lateral loads through lateral compression.

However, due to the isotropic nature of rubber, decoupling vertical and lateral stiffness in the design remains challenging [93]. Fig. 13(a) illustrates a typical floating rail fastener. Its vertical stiffness within the 20–70 kN load level is 7.1 kN/mm, while the lateral stiffness with 5–25 kN is 9.6 kN/mm, yielding a vertical-to-lateral stiffness ratio of 0.74. This configuration struggles to achieve low vertical stiffness while maintaining high lateral stability. Leveraging the structural stiffness anisotropy of DP-CTM, the rubber material inside the lateral shoulder was replaced with a DP-CTM structure (antisymmetric configuration), as shown in Fig. 13(b–c). After the replacement, the lateral stiffness increased to 14.1 kN/mm (46.9 % increase), the vertical stiffness rose to 8.6 kN/mm (21.1 % increase), and the vertical-to-lateral stiffness ratio reached 0.61. The DP-CTM structure achieved clear decoupling of vertical and lateral stiffness. This ensures both the preservation of track geometry on curved sections and the realization of vertical vibration reduction.

Fixed frogs are widely used in metro lines due to their simple structure, high reliability, low cost and ease of maintenance [94]. However, the inherent gap in the frog inevitably induces wheel-rail impact, significantly increasing localized vibration and noise [95], thereby accelerating chipping, fracturing, and uneven wear of the frog point and wing rails. Building upon this, Lei et al. proposed a wheel-rail

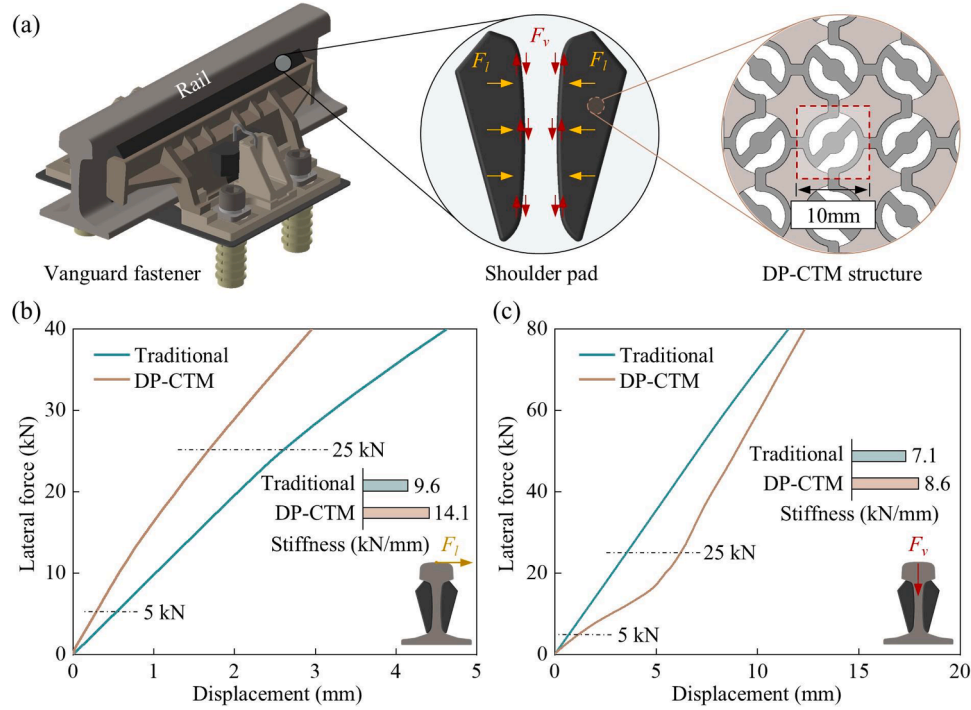


Fig. 13. Integration of the DP-CTM structure into metro vanguard fasteners and comparison with traditional designs. (a) Schematic of the vanguard fastener incorporating a DP-CTM shoulder pad, illustrating the load transfer mechanism and local DP-CTM configuration; (b) Lateral force-displacement responses of traditional and DP-CTM-based fasteners; (c) Vertical force-displacement responses of traditional and DP-CTM-based fasteners.

impact suppression device for frogs, achieving a smooth transition of wheel-rail forces in the gap in the frog [96]. This device features an upper layer of manganese steel as the impact layer and a lower layer of rubber material as the buffer layer, as shown in Fig. 14(a). However, field applications revealed that under repeated wheel-rail impact loads, the upper layer thin structure was prone to fatigue fracture. Increasing rubber hardness, meanwhile, compromised vibration reduction

performance. Thus, an effective solution integrating load-bearing and vibration reduction within limited space was urgently needed. This led to the introduction of the DP-CTM (Directly Symmetrical Configuration) structure, which enhances stiffness while improving vibration reduction performance.

Fig. 14(b) compares the wheel-rail force fluctuations at a gap in the frog and in the wing rails using the conventional rubber material versus

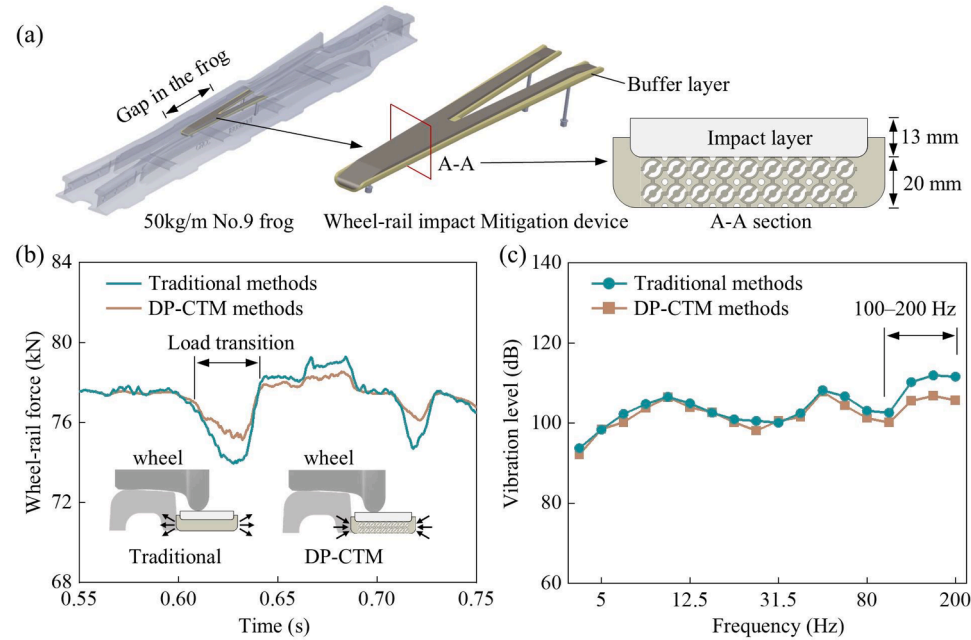


Fig. 14. Application of a DP-CTM-based mitigation device for wheel-rail impact control at metro turnout. (a) Schematic of the wheel-rail impact mitigation device incorporating a DP-CTM layer at the frog gap, and the corresponding A-A cross-section; (b) Comparison of wheel-rail impact forces during load transition for traditional and DP-CTM-based designs; (c) Comparison of vibration levels in the frequency range of 1–200 Hz for traditional and DP-CTM-based designs.

the DP-CTM structure. The higher stiffness of the DP-CTM structure, arising from the CT behavior and negative Poisson's ratio, provides stronger wheel support and results in smoother wheel–rail forces at the load transition zone. This effectively mitigates the impact effects on the frog center and wing rails. Fig. 14(c) contrasts the differences in frog vibration acceleration levels between the conventional method and the DP-CTM structure. Due to the overall smoother wheel–rail forces, the average frog vibration level decreases by approximately 1 dB ($a_{ref} = 10^{-6} \text{ m/s}^2$). Simultaneously, the resonance energy dissipation mechanism of DP-CTM reduces vibrations in the 100–200 Hz range by approximately 5 dB.

6. Conclusions

This study proposes a DP-CTM for the integrated realization of load-bearing and vibration mitigation in metro track systems. Through CT coupling, vertical loads are transformed into torsion-dominated deformation, which, combined with viscoelastic dissipation in the external TPU layer, enables energy attenuation while preserving structural stiffness. Sequential surrogate-based optimization is employed to determine the optimal geometric parameters, and theoretical as well as experimental analyses clarify the underlying mechanisms for different symmetry configurations. The main findings are summarized as follows.

The integration of the MSP+RMS sampling strategy within the sequential surrogate modeling framework enables an effective balance between global exploration and local accuracy, resulting in high-precision surrogate construction under limited sample conditions. Coupled with the MOOAA method, this approach identifies optimal geometric parameters that reconcile load-bearing capacity and vibration mitigation requirements. DOE analysis indicates that the outer diameter and wall thickness primarily govern compression-induced torsional stiffness evolution and structural load-bearing performance, whereas structural angle, inner diameter, and TPU width predominantly influence mechanical energy conversion efficiency and vibration attenuation capability.

Energy absorption in the DP-CTM is governed by topology-induced CT deformation conversion coupled with constrained viscoelastic shear dissipation in the elastic layer. Symmetry configuration significantly influences deformation pathways and internal contact sequences, with differences becoming pronounced when displacement exceeds 1.8

mm. Under identical loading conditions, the antisymmetric configuration exhibits one-sided torsion-dominated deformation, leading to a lower shear–torsion ratio and enhanced energy dissipation. In contrast, the symmetric configuration undergoes inward aggregation accompanied by a pronounced negative Poisson's ratio effect (from -0.38 to -0.63 and then to -0.167), resulting in higher load-bearing capacity and strong stiffness anisotropy. The symmetric structure achieves a minimum transverse-to-vertical stiffness ratio of 0.06, whereas the antisymmetric structure demonstrates superior vibration attenuation, with significant transmission loss at 103 Hz, 113 Hz, and 166 Hz.

This study pioneers the application of the DP-CTM concept in metro track, enabling an integrated design that combines load-bearing and vibration reduction. Despite increased structural complexity and potential cost, its superior vibration attenuation and stable load-bearing performance indicate strong application potential. The proposed concept provides a foundation for next-generation rail vibration control solutions integrating structural support and energy dissipation within a unified topology.

CRediT authorship contribution statement

Jiaxin Lei: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation. **Duoja Shi:** Validation, Software, Resources. **Mingjing Geng:** Methodology, Data curation. **Caiyou Zhao:** Project administration, Methodology, Investigation, Funding acquisition. **Weiun Liu:** Supervision, Software, Resources. **Ping Wang:** Software, Methodology, Investigation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.ijmecsci.2026.111498](https://doi.org/10.1016/j.ijmecsci.2026.111498).

Appendix A. Power flow transmission loss and vibration level drop

This appendix compares power flow transmission loss with vibration level reduction as performance indices for vibration reduction systems. Their formulations and physical interpretations are examined, and the superiority of power flow transmission loss in representing energy transmission mechanisms is demonstrated.

The vibration level drop is defined as 20 times the common logarithm of the ratio between the root mean square (RMS) value of the vibration response of the isolated equipment and the corresponding RMS value of the foundation response [97]. Here, the acceleration vibration level drop metric is selected for comparison with the power flow transmission loss, whose calculation formula is:

$$L_D = 20 \lg \left| \frac{A_1/A_0}{A_2/A_0} \right|, \quad (A1)$$

where A_1 represents the acceleration above the isolator, A_2 represents the acceleration below the isolator, and $A_0 = 10^{-6} \text{ m/s}^2$ denotes the reference acceleration.

From the above equation, it is evident that L_D depends on the impedance characteristics of the isolator and foundation. For a single-degree-of-freedom isolation system, if the effect of point apparent mass is neglected, it can be considered that $D_p = L_D$. Since the vibration level drop fails to fully reflect the actual vibration energy transfer process, neglecting apparent mass also disregards the dynamic transfer relationship. Fig. A1 compares the differences between the power flow transmission loss and the vibration level drop. The overall trends of the two curves are largely consistent. Since the power flow transmission loss incorporates the apparent mass terms A_F and A_F^B , it can reveal the complex dynamic transmission behavior within the 150–250 Hz range. Analyzing the vibration transmission patterns of isolation systems from the perspective of energy transfer, the power flow transmission loss provides a more rational approach compared to the traditional methods such as force transmission rate or vibration level drop.

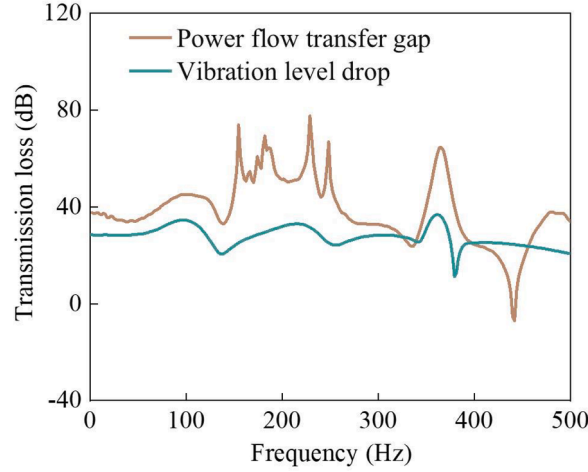


Fig. A1. Comparison of power flow transmission loss and vibration level drop in the frequency range of 1–500 Hz.

Appendix B. Universal Kriging formulation and implementation details

This appendix presents the detailed mathematical formulation and implementation aspects of the UK model, complementing the concise description provided in the main text. For a given set of sampling data, the weights for the predicted point $\mathbf{X}$ are $w(\mathbf{X}) = [w_1(\mathbf{X}), w_2(\mathbf{X}), \dots, w_n(\mathbf{X})]^T$ and $w(\mathbf{X})$ are determined by solving the following:

$$\begin{cases} \sum_{i=1}^n w_i(\mathbf{X}) b_k^*(\mathbf{X}_i) = b_k^*(\mathbf{X}) \\ \sum_{i=1}^n w_i \text{cov}(Z(\mathbf{X}_i), Z(\mathbf{X}_j)) + \sum_{k=0}^K \lambda_k b_k^*(\mathbf{X}_j) = \text{cov}(Z(\mathbf{X}), Z(\mathbf{X}_j)) \end{cases}, \quad (\text{B1})$$

where $\lambda(\mathbf{X}) = [\lambda_0(\mathbf{X}), \lambda_1(\mathbf{X}), \dots, \lambda_K(\mathbf{X})]^T$ is the Lagrange multiplier, and the basis functions $b_k^*(\mathbf{X})$ are defined as:

$$b_k^*(\mathbf{X}) = \delta_k \mathbf{X}^k, \quad (\text{B2})$$

where δ_k is a constant equal to 0 or 1.

The covariance in Eq. (B1) is defined using the Gaussian covariance function as follows:

$$\text{cov}(Z(\mathbf{X}_i), Z(\mathbf{X}_j)) = \sigma^2 R(\mathbf{X}_i, \mathbf{X}_j, \boldsymbol{\theta}), \quad (\text{B3})$$

where σ^2 represents the variance of the sampling points, and $\boldsymbol{\theta} = [\theta_1, \theta_2, \dots, \theta_D]^T$ are the correlation coefficients obtained from a maximum likelihood estimation. As shown in Eq. (B2), the number of terms in a basis function increases sharply with its order, leading to a dramatic increase in the complexity of the Gaussians model.

The basis function in the UK model is a polynomial whose order is constrained: the number of terms in the candidate basis function cannot exceed the number of sampling points used to generate the kriging matrix. Therefore, in the basis functions of the UK model, the maximum possible order K is constrained as follows:

$$C_{K+D}^K = \frac{(K+D)}{KD} < N, \quad (\text{B4})$$

where N represents the number of sampling points, and D represents the number of design variables.

Appendix C. Convergence criteria in adaptive point addition

This appendix compares the convergence performance and modeling accuracy of MSP, RMS, and combined MSP+RMS sampling strategies. Optimization trajectories, sample distributions, and surrogate model R^2 metrics are analyzed to demonstrate the enhanced global–local balance and predictive capability of the MSP+RMS approach.

A systematic comparison is conducted among the three sampling strategies under identical surrogate model construction and optimization settings, with differences arising solely from the sampling mechanisms. After 30 iterative sampling cycles based on 40 initial sample points, the resulting sample distributions and convergence trajectories are presented in Fig. C1(a–b). Regarding sample point distribution patterns, the MSP method converged fastest but risked local optima, while the RMS method insufficiently focused on extreme points. The MSP+RMS approach balanced convergence and exploration. Comparing the convergence trajectories, the MSP+RMS approach achieves the optimal objective function in fewer iterations. This method is suitable for complex modeling tasks requiring both global search and local precision.

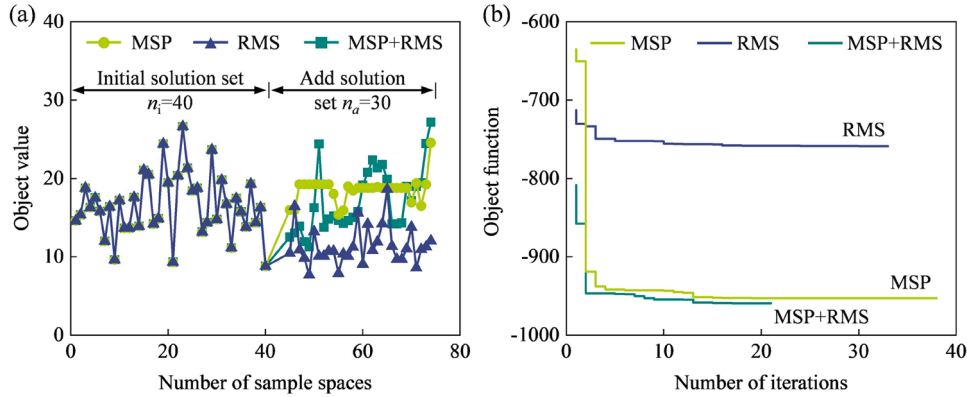


Fig. C1. Comparison of optimization performance under different strategies. (a) Evolution of objective values with increasing sample points for the MSP, RMS, and combined MSP+RMS strategies, showing the initial and added solution sets; (b) Convergence histories of the objective function with respect to the number of function and gradient evaluations for different strategies.

The coefficient of determination R^2 is commonly used to evaluate the goodness-of-fit of surrogate models. Fig. C2 displays the R^2 values for the surrogate model errors under different strategies, with cross-validation employed as the verification method and 20 sample groups used. As shown in Fig. C2(a–f), due to the relatively good linearity of displacement u , the overall coefficient of determination is favorable. The MSP and RMS methods exhibit lower prediction accuracy for power flow transmission losses D_p , with coefficients of determination of 0.41202 and 0.18188, respectively. However, by adopting the combined MSP+RMS approach, the proxy model is constructed from both local and global perspectives, resulting in coefficients of determination exceeding 0.9 for both u and D_p .

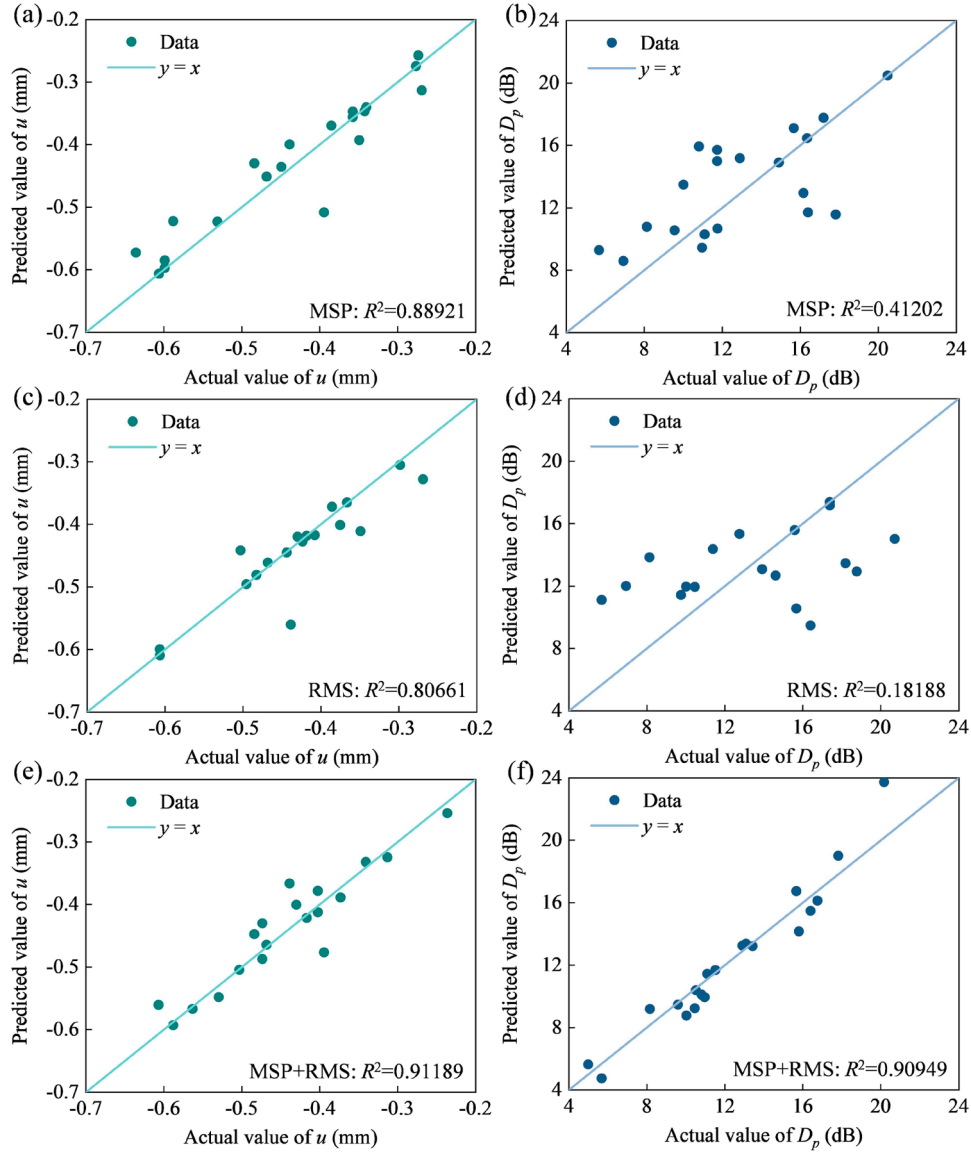


Fig. C2. Prediction accuracy of surrogate models constructed using different strategies. (a, b) Comparison between predicted and actual values of u and D_p for surrogate models built with the MSP strategies; (c, d) Comparison between predicted and actual values of u and D_p for surrogate models built with the RMS strategies; (e, f) Comparison between predicted and actual values of u and D_p for surrogate models built with the MSP+RMS strategies.

Appendix D. Geometric refinement and verification

This appendix documents the geometric refinement strategy adopted to mitigate stress concentration and verifies its influence on the global mechanical response of the DP-CTM structure.

Based on the S_3 -shaped unit cell structure depicted in Fig. 6, a cubic finite element model with overall dimensions $3 \text{ cm} \times 3 \text{ cm} \times 3 \text{ cm}$ was constructed. Depending on the symmetry, it was categorized into antisymmetric and symmetric structures. Sharp corners in the structure tend to generate significant stress concentrations under loading, which may compromise its mechanical performance and durability. By introducing a localized chamfering design, stress at sharp corners is redistributed, reducing stress concentration and improving structural stability. Simultaneously, to further increase the compressive deformation space of the TPU material, localized openings were designed on the outer structure to prevent the skeleton from bearing excessive localized loads. The specific design is shown in Fig. D1(a). Notably, despite optimized stress distribution, the structure macroscopic mechanical response remains stable. As shown in the displacement-load curve in Fig. D1(b-c), the chamfer design does not significantly affect the structure overall stiffness or load-bearing capacity, with total stiffness variation within 0.2 kN/mm . The chamfering primarily redistributes stresses in localized high-stress zones without altering the structure dominant macroscopic deformation mechanism or load transfer pathways.

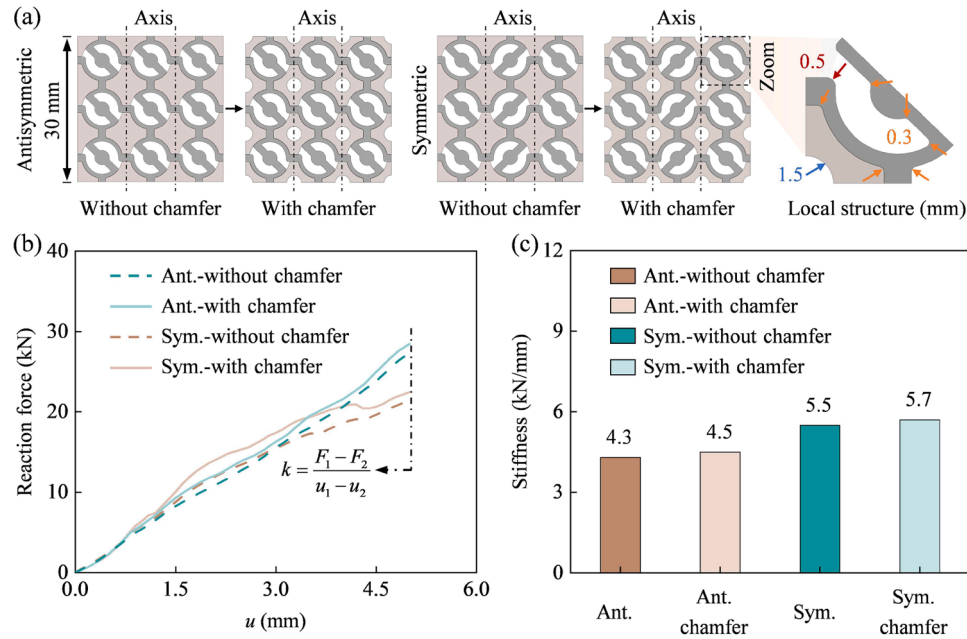


Fig. D1. Effect of chamfer on the mechanical response of symmetric and antisymmetric DP-CTM structures. (a) Geometric configurations of antisymmetric and symmetric DP-CTM layouts, together with the local chamfered structure and corresponding dimensional parameters; (b) Reaction force–displacement responses of DP-CTM structures with and without chamfer under compression; (c) Comparison of the equivalent stiffness k for different configurations.

Data availability

The authors do not have permission to share data.

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